Pixel-Based Crop Classification

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Problem Statement

Most of the food we eat comes from large scale agriculture, so it is naturally very important to understand what exactly is being grown where. Knowing what crops are currently being grown over a wide area allows us to predict and regulate markets to prevent shortages, while also being aware of potentially devastating blights due to overreliance on a specific source. Knowing where each crop is also allows us to prevent and respond quickly to disasters. It is extremely difficult to coordinate every farm in the country enough to get accurate and up to date information, so the USDA has become interested in developing a way to automatically determine what crop a plot of land contains based on satellite imagery. The result of this research is a map covering the whole of the United States at a 30x30 meter resolution describing what (if any) crop is being grown. This is called the cropland data layer (CDL). For our project, we will build a similar classifier based on data from Washington state.

Dataset Description

The data we will use to predict will come from the Landsat imagery system. Images are available from EarthExplorer in a variety of formats. Landsat 8/9 records nine spectral bands and two thermal bands with different resolutions. EarthExplorer has preprocessed images with each of these bands downsampled to a constant 30m resolution. This preprocessing also includes basic handling of cloud coverage, which will allow us to drop occluded pixels. We will pick images from a chosen month in spring of 2025 to avoid confusion from crop rotation. This will also allow the model to create more up to date predictions, as it will only require one set of images.

We will use only the nine spectral bands for our dataset. These images are very large, so we will pick a couple areas throughout the state to focus on. We will include as many images as is reasonable considering the file sizes.

This should be sufficient information to predict the correct label, but we may also try adding information derived from the elevation, such as aspect or slope. This can be sourced from the SRTM dataset (also from EarthExplorer).

Our ground truth labels are from the Washington State Department of Agriculture. This dataset is updated to 2025 and covers most of the state with 19 classes. It is provided as a vector data layer, which we will rasterize and overlay with the landsat data using ArcGIS Pro (available through Northeastern). We can then export it to a tabular dataset where each row represents a pixel. It will have at least ten columns, one of which will be a label.

Proposed Methodology

Many projects using pixel based classification find that the data is well represented by decision tree based models. We will test several such algorithms, including a single tree, a random forest, and a gradient boosted forest. We will also test logistic regression and a naive Bayes classifier.

Before training, we will split some of the data into a test set to be used to evaluate each model. We will split the data spatially instead of randomly to prevent bias from the models having trained on pixels adjacent to the test pixels.

Our data will be extremely unbalanced, since certain crops are naturally much more common than other crops. Even so, the least common class still covers over 40,000 acres, so we can downsample each of the larger classes while still having more than enough data to train on. The testing set will be left unbalanced to ensure the test statistics are not overoptimistic. We will also evaluate using the macro-F1 score and a confusion matrix of our model when comparing the model against the real solutions, as those metrics are resistant to imbalance. It should be noted that while F1 score is a better evaluation of the model's ability to distinguish classes, the actual usefulness of the model is determined by its raw accuracy against a realistically distributed test dataset. These evaluation metrics will be compared against a majority class predictor and a simple linear model. The increase in accuracy from these baselines will describe the efficacy of each model.

For each of the models, we will run a series of grid searches over increasingly narrow ranges to determine the best hyperparameters based on the F1 score of each iteration.

In addition to directly testing various models, we will also explore feature manipulation using PCA and common hyperspectral indices like NDVI. It may also be interesting to look at the results of clustering the data and see if the groups follow logical patterns based on the types of plant. To evaluate the clusters, we will consider silhouette score, Davies-Bouldin Index, and Adjusted Rand Index.

Expected Challenges + Mitigation

This project involves a significant amount of data acquisition and processing. We will need to account for the large amount of data by manually picking smaller study areas throughout Washington state. This will ensure the various analyses we run on the data are possible to compute in a reasonable amount of time.

Timeline + Task Distribution

	Alex	Jordan	Kaycie
Oct 24 - Nov 5	Obtain datasets and manipulate into usable format	Initial research into what methods to use	Review cleaned data + contribute to initial research and methodology

Nov 5 - Nov 15	Implement and test feature augmentations	Begin testing/analyzing ML models against the data	Start feature engineering and recipe building for ML models
Nov 15 - Dec 1	Analyze results and begin the final report	Analyze results and begin the final report	Analyze results and begin the final report
Dec 1 - Dec 12	Finalize report (will split up sections once available)	Finalize report (will split up sections once available)	Finalize report (will split up sections once available)

References

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